



Module 3: Transferring your data to & from CU Boulder Research Computing

Website: <https://www.colorado.edu/rc>

Documentation: <https://curc.readthedocs.io>

Helpdesk: rc-help@colorado.edu



Learning Objectives

- Ways to access your data
- Filesystem structure of CURC resources
- Data transfer with OnDemand “Files” application
- Data transfer using Globus
 - Web-based interface
 - Local Globus endpoint
 - Connecting local and CURC endpoints

Accessing Data on CURC Resources

- When you use CURC resources, the data is not on your local machine.
- Ways to access/transfer the data from/to your local machine:
 - Command line (a variety of tools – we won't cover today)
 - <https://curc.readthedocs.io/en/latest/compute/data-transfer.html>
 - **Open OnDemand (straightforward GUI) – files < 1 GB**
 - **Globus (GUI with some set up required) – any file size**

General Filesystem Structure

/home (2GB)

✓ Small important data ✓ Backed up frequently ✗ Not for sharing files or job output		
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PetaLibrary is also available via Globus!

Filesystem documentation: <https://curc.readthedocs.io/en/latest/compute/filesystems.html>

General Filesystem Structure

/home (2GB)

- ✓ Small important data
- ✓ Backed up frequently

✗ Not for sharing files or job output

/projects (250GB)

- ✓ Medium-sized important data
- ✓ Software
- ✓ Can be shared with others
- ✓ Backed up, but less frequently

✗ Not for job output

PetaLibrary is also available via Globus!

Filesystem documentation: <https://curc.readthedocs.io/en/latest/compute/filesystems.html>

General Filesystem Structure

/home (2GB)

- ✓ Small important data
- ✓ Backed up frequently

✗ Not for sharing files or job output

/projects (250GB)

- ✓ Medium-sized important data
- ✓ Software
- ✓ Can be shared with others
- ✓ Backed up, but less frequently

✗ Not for job output

/scratch/alpine (10TB)

- ✓ Large data
- ✓ Can be shared with others
- ✓ Fast Data transfer to compute nodes

✗ Not backed up!
✗ Purged after 90 days!

PetaLibrary is also available via Globus!

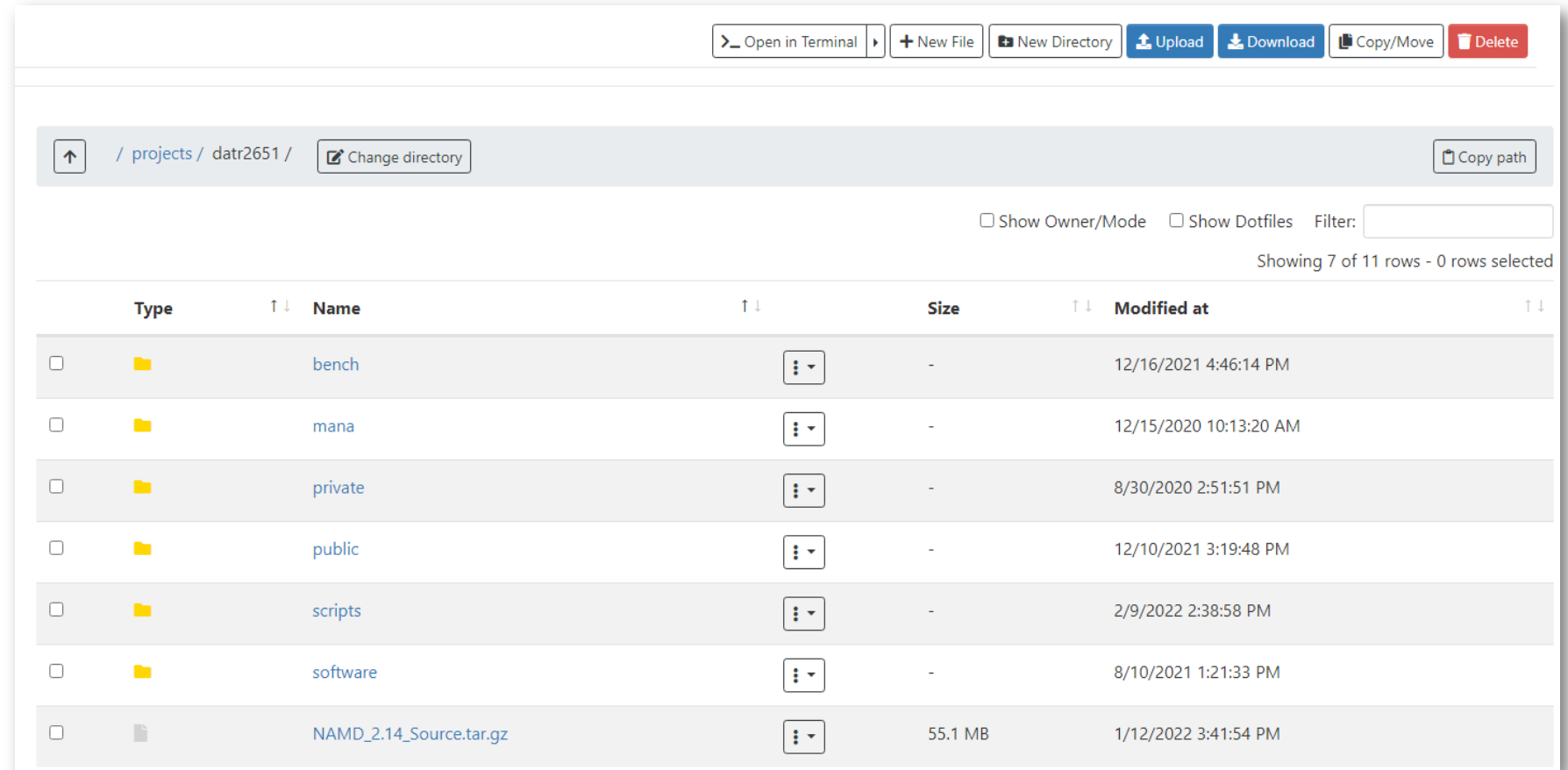
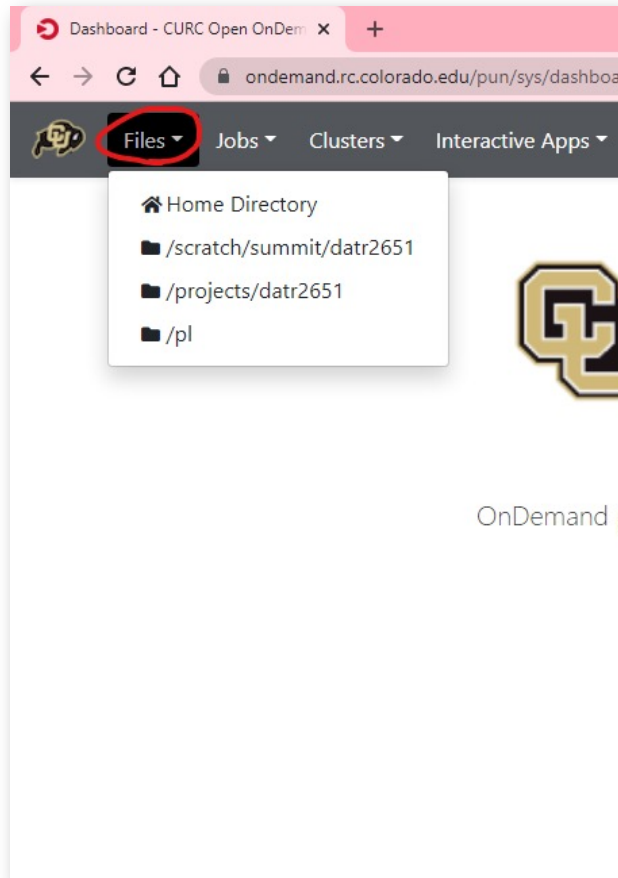
Filesystem documentation: <https://curc.readthedocs.io/en/latest/compute/filesystems.html>

GUI option - Open OnDemand

- No command line required!
 - <https://ondemand.rc.colorado.edu/>
 - <https://ondemand-rmacc.rc.colorado.edu/>
- File management
 - Create, Delete, Move, and Rename
- File transfers
 - Upload and Download



Open OnDemand “Files” Interface



GUI option - Globus

Globus is a service that allows for users to reliably move, share, and discover data

- Command line also available
- Our recommended way to transfer data
 - Stable and fast data transfers
 - Transfers continue if a user disconnects
 - Web GUI or Globus Connect Personal GUI
 - Works well with cloud storage providers



Globus Demo

- Logging into the web-based interface for Globus
 - Provides a nice GUI for managing files on CURC resources
- Installing a Globus Endpoint on your local machine
 - Allows you to interact with files on your local machine via Globus

Globus Login to Web-based interface

1. Navigate to <https://app.globus.org>
 - CU Boulder users
 - Select “University of Colorado at Boulder” in the dropdown menu
 - CSU users
 - Select “Colorado State University”
 - AMC and RMACC users
 - Select “ACCESS CI (formerly XSEDE)”
2. Login with your credentials
3. Continue with onscreen prompts until you are brought to the Globus Web GUI

Globus Web GUI

Let's take a look!

Local Globus Endpoint

- Required if you want to transfer data between your machine and CURC resources
- Navigate to <https://www.globus.org/globus-connect-personal>

Install Globus Connect Personal

Create a Globus collection on your laptop. Globus Connect Personal is available for all major operating systems.



Globus Connect Personal for Mac

Mac OS X 10.9 or higher

INSTALL NOW >



Globus Connect Personal for Windows

currently supported Windows versions

INSTALL NOW >



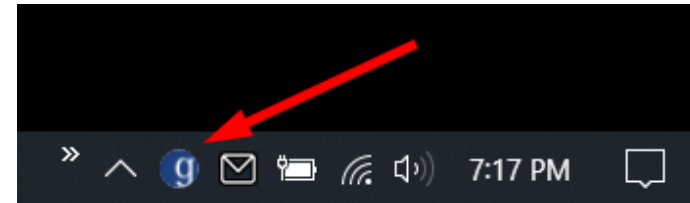
Globus Connect Personal for Linux

for common x86 distributions

INSTALL NOW >

Local Globus Endpoint - Installed

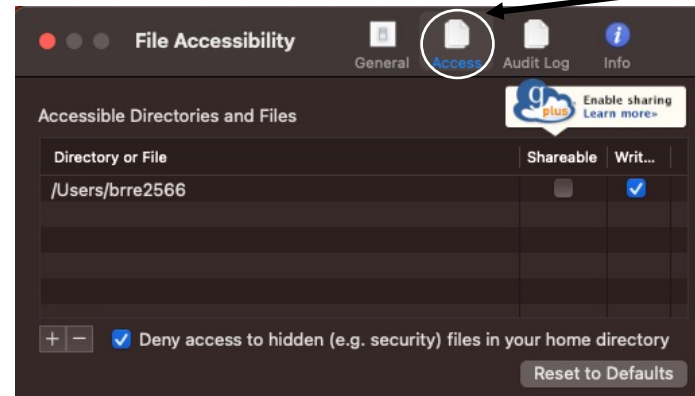
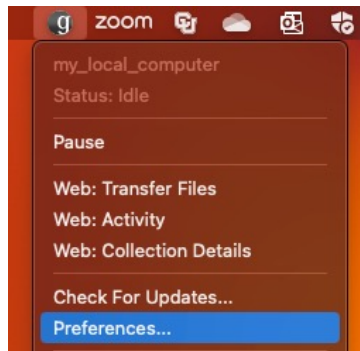
- For “Collection Name” you can use anything name you would like
 - You will use this as a reference name later, so please remember it!
- For Mac and Windows, it is easy to see that Globus has been installed:



- For Linux, at the end of the installation, the setup will exit with a message stating it was successfully set up, and the main Globus Connect Personal application will launch

Local Globus Endpoint - Configure

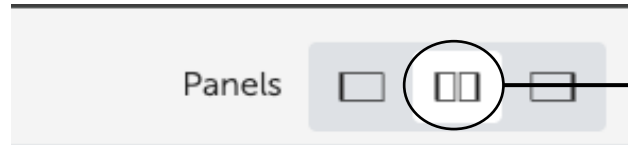
- Once the Globus endpoint has been installed on your computer, you may need to configure it.
- During Configuration you can let Globus know what directories or files you want it to access.



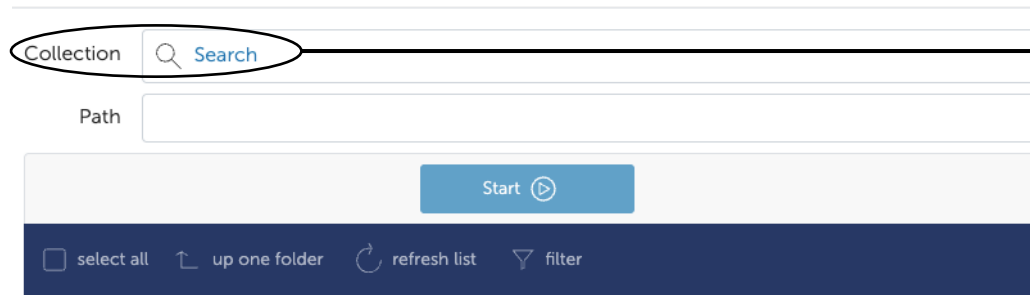
Select the “Access” tab and add folders you want to be accessible by Globus

Connecting local endpoint to CURC resources

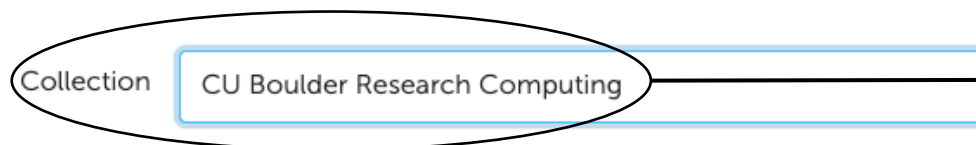
Globus



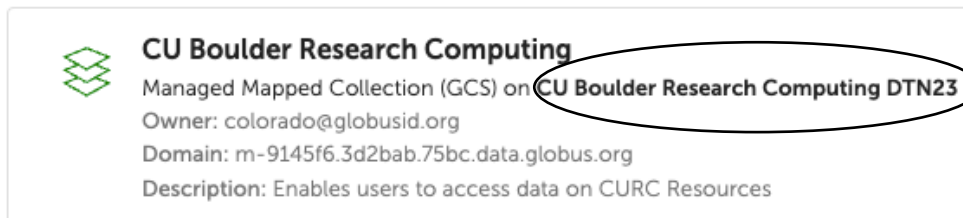
Step 1: Select “set two pane”
in the upper right corner



Step 2: In the left pane
click the search bar

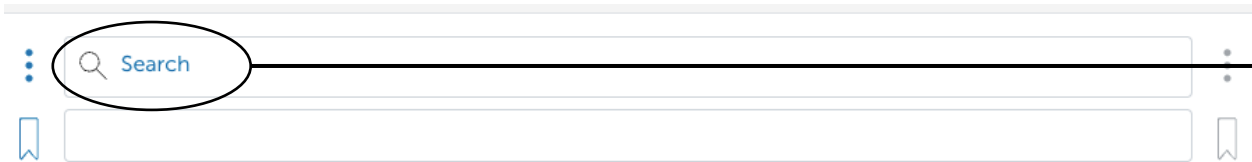


Step 3: Search “CU Boulder
Research Computing”

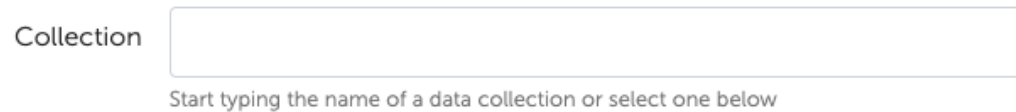


Make sure you are
using DTN23!

Globus



Step 4: In the right pane, click on “Search”



Step 5: Select the “Your Collections” tab

Step 6: Click the collection name that **you** chose when you created your local Globus endpoint

Advanced topic: Globus Shared Endpoints

- Globus offers 'shared endpoints', which allow you to share your data with external collaborators (i.e. you can share data with people who don't have CURC accounts)
- CURC provides this capability, however, it is only available for PetaLibrary customers
- Shared Endpoints generate a shared collection that can be accessed with a link
 - For an example, see <https://scholar.colorado.edu/concern/datasets/9593tw13k>
 - You can assign various permissions to specific users or all users within Globus
 - More information is provided at <https://docs.globus.org/how-to/share-files/>

Review of what we have covered

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Any Questions?